

Task 12.3 Order Fulfilment Analysis Report

Objective of the Dashboard

The purpose of this dashboard is to analyze **order fulfilment performance** by evaluating:

- Delivery efficiency
 - Shipping performance
 - Delay and cancellation impact
 - Business impact (sales & profit)
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KPI Summary

Business Questions Answered:

- What is the overall order volume?
- How much revenue and profit is generated?
- How efficient is the delivery process?

Total Orders	Total Sales	Total Profit	On Time Orders	Late Orders	Early Orders	Cancelled Orders
113K	23.16M	2.61M	46K	62K	26K	5K

KPIs Used:

- Total Orders
 - Total Sales
 - Total Profit
 - On-Time Orders
 - Late Orders
 - Early Orders
 - Cancelled Orders
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Key Insight:

- Late orders (62K) exceed on-time orders (46K), indicating **serious fulfilment inefficiency**.
 - A significant portion of orders are not meeting expected delivery timelines.
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Chart 1: Order Distribution by Delivery Status (Donut Chart)

Business Question:

How are orders distributed across delivery outcomes?

Insight:

- Majority of orders fall under **Late Delivery**, showing operational delays.
 - Early and on-time deliveries are comparatively lower.
 - Cancelled orders represent a smaller but critical failure segment.
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Chart 2: Total Sales by Delivery Status (Treemap)

Business Question:

How is revenue distributed across delivery performance?

Insight:

- A large portion of sales is associated with **late deliveries**.
 - Indicates that even high-revenue orders are not being delivered efficiently.
 - Suggests potential **customer dissatisfaction despite strong sales**.
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Chart 3: Total Profit by Delivery Group (Treemap)

Business Question:

What is the profit impact of delivery performance?

Insight:

- **Delayed orders generate the highest profit**, which is a critical finding.
 - Indicates:
 - High-volume orders are delayed
 - Operational inefficiencies exist despite profitability
 - Opportunity to improve efficiency without losing revenue.
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Chart 4: Total Orders by Delivery Group (Bar Chart)

Business Question:

How many orders fall into success vs failure categories?

Insight:

- Delayed orders dominate the distribution.
 - On-time deliveries are significantly lower.
 - Cancelled orders, though smaller, indicate process failure.
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Chart 5: Orders by Shipping Type & Delivery Group (Stacked Bar Chart)

Business Question:

Which shipping methods contribute to delays and cancellations?

Insight:

- **Standard Class shipping contributes the highest number of delayed orders.**
 - Faster shipping methods (First Class, Same Day) show better performance.
 - Indicates that **shipping method is a major root cause of delays.**
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Slicers (Filters)

Slicers Used:

- Order Date
 - Shipping Type
 - Customer Segment
 - Customer State
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Business Questions:

How does performance vary across:

- Time?
 - Regions?
 - Customer segments?
 - Shipping methods?
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Insight:

- Enables dynamic analysis:
 - Identify delay trends over time
 - Detect regional inefficiencies
 - Compare customer segments
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Overall Key Insights

1. High Delay Rate

- Delayed orders are the largest category
 - Indicates **logistics inefficiency**
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2. Revenue vs Performance Gap

- High sales and profit are linked to delayed deliveries
 - Suggests **volume is high but execution is weak**
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3. Shipping Type Issue

- Standard shipping is the main contributor to delays
 - Opportunity to optimize shipping strategy
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4. Cancellation Impact

- Though smaller in volume, cancellations directly impact revenue and customer trust
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10. Business Recommendations

1. **Optimize Standard Shipping**
 - Improve logistics or reduce dependency
 2. **Focus on On-Time Delivery**
 - Increase operational efficiency
 3. **Reduce Delays Without Losing Volume**
 - Maintain high sales but improve execution
 4. **Monitor High-Risk Regions**
 - Use slicers to identify problematic areas
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11. Conclusion

This dashboard successfully highlights:

- Operational inefficiencies in delivery
- The financial impact of delays
- Root causes through shipping analysis

It enables stakeholders to:

- Make data-driven decisions
- Improve delivery performance
- Enhance customer satisfaction